

Front-Ends and
Apple Computer in
the
1990s

A Report to the Management of Apple Computer USA

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This is the final report of a study conducted by Aberdeen Group, Inc. for Apple Computer USA Marketing. Aberdeen was asked to examine the market for clientserver computing (also called cooperative processing), with an emphasis on terminal front-ends to existing mainframe applications (sometimes called non-invasive frontending or frontware).

We conducted eleven interviews of large mainframe-size sites in three U.S. geographic regions to determine current attitudes and plans for cooperative processing, and attitudes about Apple Computer and its competitors. The interviews included cooperative processing decision makers in the banking, brokerage, government, insurance, retail, transportation, and utility industries.

The Executive Summary section contains an overview of the findings of our study, along with Aberdeen Group's conclusions and recommendations.

The Detailed Findings, Conclusions and Recommendations, section two, contains our detailed research findings, conclusions for Apple based on the findings, and recommendations for Apple's marketing programs. This section also contains Aberdeen Group's analysis and recommendations regarding Apple's U.S. marketing programs aimed at improving market share at large, mainstream commercial accounts.

Section three, the Frontware Applications section, discusses a number of real and hypothetical uses of non-invasive front-ends. These examples are designed to enhance Apple's understanding of practical uses for frontware.

The Interview Contact Summaries, section four, contain the complete record of the inperson interviews conducted at eleven mainframe sites. This material weighs heavily in our research findings and should be read by marketing staff for insights into the current thinking and attitudes of a cross-section -- but not a scientific sample -- of mainframe buyers and recommenders of cooperative processing technology.

Appendix A contains the slide presentation and script used during the interviews to discuss the road to cooperative processing and Apple's role in front-ending.

I. EXECUTIVE SUMMARY The Market for Frontware and Client-Server Computing

The market for non-invasive front-ends to mainframe hosts (frontware) is small and mixed in buyer's minds with the also-emerging move towards cooperative processing using the client-server model. Aberdeen defines cooperative processing as desktop client-based display, decision support, transaction validation and communications to a server.

Buyers are aware of frontware, and intend to evaluate it -- along with potential cooperative processing client platforms -- as early as this year. However, the market is immature and buyers do not have the background or experience to weigh the merits of frontware supplier claims, particularly technical evaluations.

We believe the frontware pilot application jump-start program recently announced by Apple is a necessary marketing program which will obviate drawn-out technical sales campaigns while positioning

Apple for early revenues from production application volumes at successful pilot sites. However, we are concerned Apple is not setting its sights high enough by planning less than 25 pilots in 1991. The Fortune 1000 needs a 1991 wake-up call that clearly indicates Apple is ready, willing, and able to compete for enterprise commercial workstation business.

Our research shows no vertical industry bias. We conclude the market for frontware applications is cross-industry and its applicability is enterprise-dependent. We suggest broad-based marketing programs such as seminars in order to reach a wide audience with Apple's frontware capabilities messages.

The cooperative processing market is likely to mature only a year or two after frontware does in 1993. We conclude that the frontware market and the cooperative processing market are maturing in parallel. We see enterprises choosing frontware for existing applications which need little changing, and the client-server model for new applications or re-engineered applications. We suggest that Apple market its frontware capabilities in conjunction with its cooperative processing capabilities as it is likely enterprises will see suppliers who can provide both.

As our research indicates many major enterprises are on a path towards evaluating and selecting cooperative processing -- and frontware -- workstations over the next eighteen months. We believe it is imperative that Apple position itself to be considered for both frontware and client workstations because these workstation decisions by MIS will establish buying patterns for the rest of the 1990s.

The MIS View of Apple

Over eighty percent of corporate America is committed to and generally satisfied with the IBM PC. This group appears unwilling to consider Apple for future client-server and front-end projects, and is likely to choose the IBM PC architecture for future applications. We see substantial corporate inertia surrounding the IBM PC, and this should not be a surprise to Apple Computer.

Our research shows no clear attributes to organizations which have chosen or might choose Apple. This market fragmentation makes it difficult to target marketing programs.

We are concerned with the numerous myths and old history repeated about Apple to us. If corporate MIS really believes that Macintosh is a toy computer, that it has no substantial business applications available, and does not connect to mainframes or PC-compatible LANs, then we believe Apple has some severe credibility problems which must be resolved before Apple's cooperative processing messages are even heard.

What MIS Wants From Apple Computer

Few MIS executives interviewed by Aberdeen Group had ever had a presentation to them made by Apple. They are quite ignorant about Apple's general capabilities, let alone Apple's workstation possibilities. In contrast, IBM is either on-site daily or making frequent information updates to these (potential) buyers. The clear remedy is more frequent and aggressive direct marketing activities.

Apple has a good reputation in MIS as a quality company. We believe Apple can capitalize on this reputation.

We were surprised that several executives we spoke to were surprised -- to the point of being emotional -- when they learned from us a quick overview of Apple's present capabilities as a workstation. They wanted more information, wondered why Apple wasn't calling on their enterprises, and left unstated the fact that they had never been given any choice but an IBM PC architecture because they were unaware of Apple's capabilities.

It appears Apple's media coverage is being ignored by many MIS buyers as irrelevant to them.

MIS is satisfied with buying from and being supported by the dealer channel of distribution.

At two sites where Apple had invested a significant selling, the company was rewarded with (or soon will be) significant business volume.

Other Significant Findings

With the installed base of 5,000,000 IBM 3270s declining at an estimated 15%-20% a year, a market exists for desktop PCs which replace 3270s.

Newly purchased PCs are today almost always connected to a LAN. Large MIS sites have typically invested heavily in Novell Netware, so Apple connectivity to Netware is an entry hurdle -- and many MIS buyers are unaware that Macintosh is supported by Netware 286 today.

Aberdeen anticipates a rapid growth in the market for departmental communications gateways running SNA SDLC to the IBM mainframe host. We believe Apple may have a price-per-seat advantage using Macintosh with gateways from Avatar, DEC, and Tri-Star, and we recommend Apple focus on multi-port connectivity costs and multi-session configuration issues versus IBM PC options.

PCs are purchased today through an entrenched corporate buying procedure which is unique to each enterprise. Every enterprise requires a different sales and marketing strategy.

Microsoft Windows 3.0 is the client software environment most frequently mentioned as the likely target for client applications. OS/2 is used by those requiring multithreading or committed to IBM's OfficeVision/2. Unix workstations are non-starters for corporate MIS, but Unix servers are attracting increasing interest due to their price performance advantages, particularly for RDBMS-based decision support.

As of November, MIS executives were expecting budget increases ranging from no increase up to the 5% rate of inflation.

Large sites expect to develop client-server application development expertise inhouse over time.

IBM is showing virtually all of its utilities and packages with front-ends, blessing Easel Corp. as the frontware-of-choice. IBM's SAA architecture for cooperative processing from PS/2s to CICS/DB2 is still two years away.

Software AG's Natural Connection has Macintosh support for client CASE tools as well as a 4GL for mainframe host application development. We recommend an evaluation by Apple of the potential for expanded joint marketing with SAG of the Natural Connection for Macintosh products.

II. MAJOR FINDINGS AND RECOMMENDATIONS

THE MARKET FOR FRONTWARE AND FRONT-END COMPUTING

Findings

1. The frontware market over the next three years is limited, maturing, but still unstable.
2. We size the market for frontware software at under \$50 million in 1990, growing at 20%-25% in 1991.
3. Corporate buyers are aware of frontware as a class of products.
4. Buyers are unaware of the wide range of technical capabilities offered in frontware products. Easel and InFront are most frequently mentioned.
5. IBM has been touting Easel as not only the client development toolset for future client-server applications, but also the non-invasive front-end to retrofit over existing applications.
6. Only one large existing Apple site (University of Texas) was aware that Apple with Mitem's MitemView is a viable platform for front-ending mainframes.
7. While numerous sites mentioned they had or would evaluate frontware for their enterprises, only Delta airlines had implemented an application with a standard PC (Prudential-Bache's system used proprietary CTOS machines).
8. Merrill Lynch stated they were fearful that they could never manage the host application version control rigorously enough. They worry that a mainframe screen change could crash all the frontware PCs. [Unfortunately, true client-server applications make the problem even worse].
9. There is no strong indication that frontware is a phase during the transition to client-server-based cooperative processing.
10. Aberdeen surveyed mainframe sites in the banking, brokerage, government, insurance, retail, transportation, university and utility markets. Our assumption was that the financial services industry was most likely to embrace frontware due to the competitive nature of these businesses. Our research does not support this assumption.
11. We conclude that the market for frontware applications is cross-industry and enterprise-specific.

Conclusions

Targeted vertical marketing programs are inappropriate due to the cross-industry and enterprise-specific nature of the frontware market.

Apple must use broad-based marketing programs such as

seminars to locate and educate corporate buyers.

Buyers do not understand the range of capabilities in frontware products so they are unlikely to respond positively to a technical sell.

As discussed in the next Major Finding, buyers are likely to overlap front-end computing with cooperative processing computing.

Apple is not generally recognized as a potential supplier of frontware hardware and software.

Aberdeen's conclusion is that a marketing program combining Apple's frontware and client-server capabilities is the prudent course:

- Frontware (non-invasive front-ending) is best suited to existing applications which will change little. The market for frontware will likely mature in two years -- just ahead of the market for true cooperative processing using the client-server model;

- Application re-engineering using frontware competes with new-application development using the client-server model for the same 10%-20% of the MIS budget allocated for "new application development";

- Frontware and client-server are mutually exclusive. Aberdeen believes MIS will either retrofit a mainframe application with frontware or rearchitect the application using a client-server architecture.

Implications

It is likely many buyers will confuse frontware with more advanced client-server development tools.

The recommended broad-based marketing programs are likely to be more costly and less effective than if vertical market programs could be employed. We see no alternative.

THE EVOLUTION OF THE MARKET FOR COMMERCIAL CLIENT-SERVER

COMPUTING

Findings

1. There is no headlong rush to client-server. We estimate well over 95% of potential applications at mainframe-size sites are still using dumb-terminal -- usually 3270 -- communications to the mainframe host from either 3270's or PCs using terminal emulation.
2. Most believe 1994 is the time when their organizations will be fully up to speed with client-server technology and that the client-server market reaches maturity.

3. Client-server is not for everyone nor is it a sure thing. Our research showed a fair amount of skepticism on the ultimate productivity value of client-server, concern over the costs and lack of available technology, and general agreement that data entry jobs (e.g. retail sales clerk) would benefit least from client-server using GUIbased clients such as Macintosh.
4. There are multiple accepted definitions for client-server, a source of confusion. Buyers think they know what it is, and suppliers all claim to have it. Further, frontware is likely to be mixed in with client-server in buyer's minds. Lack of market maturity is causing a failure to communicate.
5. The multiple technologies needed to support open client-server computing are not mature. Corporate buyers are seeking to mix and match PCs, LANs, communications gateways, and database servers in any and every way. The technology, particularly interoperability software, will not be generally available until early 1992 -- and perhaps later.
6. In spite of the technology uncertainties, buyers are proceeding to evaluate and select client hardware and software platforms in 1991.
7. IBM is neither leading nor following. No other hardware suppliers -- with the possible exception of Hewlett-Packard with NewWave -- are in contention to lead this market with proprietary solutions. The market will proceed with or without IBM, but will grow more quickly once IBM gets its act together.

Conclusions

Apple must move quickly to establish its presence in the market for frontware- and client-based application workstations. Even though technology barriers and slow buyer acceptance will impede market maturity until the mid-1990s, buyers are now or will shortly evaluate their options for new desktop platforms.

It is imperative that Apple position itself to be considered for both frontware and client workstations because these workstations decisions by MIS will establish buying patterns for the rest of the 1990s.

Implications

A targeted non-invasive front-end marketing program not only misses the client-server target but leaves buyers with the impression Apple can only solve part of the problem.

MIS VIEW OF APPLE COMPUTER

Findings

1. A significant portion -- over 80% -- of the main-line MIS market is strongly committed to the IBM PC architecture. The bulk of MIS market appears unwilling to even consider Apple for future client-server and front-end projects. They have already made a choice based around the IBM PC architecture.

2. MIS (Management Information Systems) is generally satisfied with the IBM PC and its likely future. We see substantial organizational inertia surrounding the IBM PC.
3. MIS is comfortable with IBM PC issues: sizing, installing, buying, upgrading, applications, local area network (LAN) connectivity, host connectivity. Apple threatens the status quo with unknowns and higher support costs.
4. Those organizations which will use Apple or would consider Apple are hard to characterize and therefore will be difficult to market to. There is severe market fragmentation.
5. MIS believes incorrectly that Apple cannot connect to standard PC networks from Novell, 3Com and Banyon. The "Apple Connects" program seems not to have connected. There is considerable outdated information and misconceptions about Apple's capabilities in the MIS environment.
6. MIS believes a lack of applications seriously hinders Apple's potential applicability for their organization. For many purchases, MIS seeks business solutions, not base platforms.
7. The MIS view of Apple is polarized by enterprise. Apple is either a recognized player by MIS, or else is totally ignored. There is little middle ground.
8. As a company, Apple enjoys a favorable reputation as a quality company during a period when MIS buyer's are seeking a flight to safety -- dealing with only the highest quality, financially strong, and reputable suppliers. Aberdeen believes Apple's quality reputation is one of its understated assets.

Conclusions

The general MIS attitude about Apple Computer should not be surprising.

The likelihood of converting a large fraction of the heathen IBM PC-oriented MIS masses to Macintosh is nil.

The mis-information about Apple connectivity, applications, and general applicability in a large MIS organization can and should be addressed with concerted marketing programs, but will take time.

Implications

Apple's market share in this decade is unlikely to exceed 15% of the business PC market, with large MIS shops putting up the largest resistance to Apple's marketing efforts.

WHAT MIS WANTS FROM APPLE COMPUTER

Findings

1. Few MIS buyers are familiar with what Apple can do today. Many believe myths and stereotypes that the Mac is a toy computer, is overpriced, and that it cannot network with PCs or mainframes using industry standards. There is considerable missing information and wrong information present about Apple within corporate MIS.
2. Few interview respondents had ever had a presentation made to them at their site by Apple. Our interviewers saw a high degree of emotional response from interviewees (Continuum and USAA) who were surprised to learn the impressive extent of Apple's capabilities. They want to be called on and to be sold to.
3. While Apple has a good corporate reputation in MIS, the business justification for Apple Macintosh vs IBM PC-compatibles is not apparent. MIS wants sound productivity and cost/benefit information.
4. Where Apple has done a significant and long-term selling job (Delta and University of Texas), the company has been rewarded by MIS management with significant business.

Conclusions

While Apple computers are sold through the dealer channel, the selling to corporate MIS needs to come from Apple's own Large Account Marketing organization.

The strong emotional response noted in some interviews indicates some corporate buyers understand the tough road to cooperative processing using the IBM platform ... and they are looking at alternatives.

Our conclusion is that Apple's product, support, and field sales issues are of minor importance. The real issue is that buyers believe myths and old truths about Apple, are unaware of the connectivity, frontware and client capabilities, and of the business applications now available on the Macintosh.

Implications

Apple's media coverage is not reaching the large-enterprise MIS buyer. These buyers can only be sold to directly and over time.

Before Apple can market its frontware and client workstation capabilities to MIS, Apple sales must overcome substantial buyer resistance hurdles centered on broad product knowledge and corporate capability issues.

ATTITUDES ABOUT APPLE COMPUTER ARE POLARIZED

Findings

1. We observed a polarized buying pattern. At 10%-20% of the surveyed sites, Apple was an established supplier with a healthy installed base. MIS was positive to enthusiastic about Apple.

2. However, the other 80% of the large enterprises we surveyed had such a low visibility on Apple that when asked how many Macintoshes were installed, replied they either didn't know, didn't care, or that "a couple were around somewhere."

Conclusions

The polarized buying patterns constrain Apple's marketing strategy options. Clearly, the easiest course is to grow usage at sites where Apple is already a recognized.

The hard -- and necessarily expensive -- course is to improve corporate America's attitude towards Apple as a viable supplier for mainstream workstations.

DISTRIBUTION CHANNELS

Findings

1. Even the largest organizations routinely buy IBM PCs from the same dealer channels, such as Businessland, that sell Apple computers today.
2. The exception is multi-thousand unit buys which are bid directly (Prudential Bache and Merrill Lynch).
3. Dealers also provide support to the large enterprise Infocenter.
4. A PC supplier such as Apple is expected by MIS to provide pre-sales information and post-sales backup support to the dealer.

Conclusions

Based on the lack of MIS interest in and knowledge about Apple, we conclude that dealers are taking orders for IBM PCs, rather than selling what's best for the customer, and not educating the market about Apple.

Implications

Dealers have few incentives to sell Apple when the customer calls and says "IBM PC". Dealers cannot be realistically expected to objectively sell Apple under these conditions.

If Apple business practices cannot provide dealer incentives to counter walk-in IBM PC business, then marketing programs conducted jointly with dealers are necessary.

TRENDS IN TERMINAL USAGE AND CONNECTIVITY

Findings

1. The installed 3270 base of about 5 million is declining at an estimated 15%-20% per year through 1993. PCs, generally IBM-compatible, are viewed as the replacement of choice.
2. The process of 3270 divestment will be gradual throughout the 1990s for two reasons. First, the capital costs involved with replacing 10-20,000 devices at large sites is staggering. Second, data entry jobs are not perceived as benefiting greatly from a PC, particularly a GUI-based PC.
3. New PCs are almost always connected in a LAN as they are installed. Older, existing PCs may be left standalone. LANs are being connected to mainframe hosts, an important precursor of mainframe-based client-server systems.
5. Large sites agree with IBM's recent conclusion that client-to-server connections are best routed through a departmental communications gateway server running SNA SDLC as a means of governing throughput and controlling the potential for unmanageable numbers of simultaneous mainframe sessions. This suggest Apple should reinforce its relationships with Avatar Digital Communications Associates and Tri-Data Systems -- or supply multiport SNA gateways directly.
6. Many mainframe MIS executives believe Apple's connectivity is limited to AppleTalk, and that the AppleTalk is not a robust enterprise-capable LAN platform. Before Apple can run with client-server, it has to walk with connectivity in buyer's minds.
7. Aberdeen research shows Novell's Netware to be the dominant LAN operating system today. The ability of Macintosh to connect to Netware 286-based servers in not generally known by MIS executives, a hinderance to Apple. Further, Novell's delay in delivering Mac connectivity to Netware 386 servers will increasingly hurt Apple as the Netware 386 installed base grows rapidly in 1991.

Conclusions

A market exists for desktop PCs which replace 3270s.

Old-style "3270 add-in board" solutions to mainframe connectivity may quickly transition to gateway server topologies.

Apple must be widely known as having excellent connectivity to Novell LANs.

Implications

Apple may have a LAN cost-of-ownership advantage using multi-port network adapters such as Avatar, DCA or Tri-Data's. This advantage, coupled with improved Netware 386 connectivity, would be favorably viewed by corporate MIS.

CORPORATE PC POLICY

Findings

1. The approval of PC purchases is made at levels ranging from individual departments all the way up to a DP Policy committee. It strictly depends on the individual enterprise.

2. The approval of PC suppliers is generally made at a high organizational level, is rarely changed, and is implemented by Purchasing. However, some organizations allow flexibility in buying IBM PC clones for individual projects.
3. Approved PC suppliers invariably include the IBM PS/2, sometimes include Apple and Compaq, and rarely includes PC clones.
4. MIS Infocenter managers say they don't want the support problems associated with clones.
5. Large enterprises tend to be more rigid than small organizations.
6. PC sales and installation are handled by dealers at all but the over-5,000-PC enterprise. Buyers seem generally pleased with dealer support and see no reason to change distribution channels, a plus for Apple.

Conclusions

Apple requires a custom sales and marketing strategy for every enterprise.

Apple will have to sell high into increasingly intransigent management to earn approval as a supplier of record.

Implications

Only concentrated marketing and selling at targeted enterprises is likely today to gain Apple the status of a general-purpose corporate PC, alongside IBM's PS/2.

USING PCs IN PRODUCTION APPLICATIONS

Findings

1. Based on user's definitions of what users want from the so-called client-server model, Aberdeen Group defines a client-server application as one where the PC client performs data entry, display, editing, and message-handling (the transaction) to a back-end database server.
2. Very few client-server applications fitting the above definition are installed today. Those that exist are mission-critical and highly customized to the enterprise's needs.
3. PC-based client-server is practically non-existent in 1990.
4. Large organizations are generally following IBM's lead in believing the clientserver model will dominate application design by 1994.
5. The acceptance of client-server is not universal. Some believe there will be little productivity benefit to data entry workers, while conceding benefits to casual knowledge workers.

6. Most expect new applications will use the client-server model -- or, at the least, will use desktop computers as the human-interface platform.
7. The majority of large organizations surveyed expect to make substantial, longterm technology decisions by the end of 1991 regarding client-server. They are considering tools now and some will implement pilot applications beginning in 1991. Thus, the market window is immediate.

Conclusions

Buyers who have active projects are choosing their tool-sets over the next year, making 1991 a milestone for decisions that will set standards for the rest of the decade. Apple should strongly consider an early program to establish the company as a leading player with deliverable client-server solutions today.

The market is immature. Buyers are selecting technology with the real deployment volumes delayed until the mid-1990s.

Client-server and cooperative processing are imprecise terms which are rapidly becoming marketing buzz words with no useful meaning.

Apple should invest heavily in 1991 marketing programs to ensure the company's client-server capabilities are understood, and to generate pilots for mid-1990s production systems.

Implications

Delaying an active client-server marketing program in 1991 may lock Apple out of a generation or more of corporate systems.

BUYER'S EVALUATION OF SOFTWARE ENVIRONMENTS

During the course of our in-person interviews, we asked for off-the-cuff comments on the following six client software environments.

Findings

Microsoft Windows 3.0 - Is regarded as the most desirable platform for client development in 1991. Most consider it adequate for the job.

Hewlett-Packard NewWave - Is not widely known, but gets support from the most technically knowledgeable as not only the elegant version of Microsoft Windows 3.0, but as having a rational approach to open server computing.

IBM OS/2 - Only those who require multiple tasks (e.g. multithreading) or are diehard fanatics for the yet-undelivered OfficeVision/2 are interested in OS/2.

SCO Open Desktop - This Unix software environment is virtually unknown by the corporate commercial application developer>

Sun Open Look - Sun is gaining commercial market share where multiple active windows are required, such as in foreign exchange trading applications. Otherwise, Open Look is a non-starter.

Apple Macintosh - Is not being seriously considered except at those sites which already have made a commitment to Apple or are being actively sold by Apple.

Conclusions

Microsoft Windows 3.0 is surely Apple's first, second, and third competitive priority.

The fact that Apple is not being actively considered or asked to bid on the long-term enterprise client hardware platform at many sites is indicative of the entrenched mentality surrounding the IBM PC.

Aberdeen continues to see almost no desire for Unix desktop systems in corporate America for use as production clients. We do see rapidly growing interest in Unix servers and in Unix (particularly Sun) workstations as software development platforms.

OUTLOOK FOR MIS BUDGETS IN 1991

Findings

1. Those surveyed who knew the entire MIS budget picture report that budgets will range between flat and up 5% -- the rate of inflation.
2. No major MIS resources will be devoted to client-server technology alone. However, key applications which benefit from the client-server model will likely move forward with the selection of base technology, software tools, and pilot-program hardware platforms in 1991.
3. MIS will develop experience in client-server through in-house projects. MIS does not expect to outsource client-server projects to avoid the learning curve. However, we believe that the industry will generally stumble in implementing pre-1995 clientserver systems. This is likely to cause a backlash as buyers react to being oversold -- yet again -- on immature technology.
4. Outsourcing application development to systems integrators such as Andersen Consulting for key new applications continues to grow. However, Aberdeen research shows that these contracts frequently use conservative -- that is, old -- development methodologies and conservative hardware platforms. We observe little client-server development work by the system integrators today, but considerable planning is underway.

Conclusions

As of November 1990, MIS was looking for a so-so budget year in 1991. There were no indications a recession is causing drastic budget cutbacks.

The complexity and immaturity of cooperative processing

technology, coupled with MIS's lack of experience with this new technology, will cause MIS to use system integrators to help jump-start their in-house efforts.

IBM IS SAYING ONE THING AND DOING ANOTHER

Findings

1. IBM is demonstrating virtually all its software products with add-on graphical front-ends. This serves to whet the market's appetite for GUI front-ends.
3. While IBM has blessed Easel as the frontware of choice, knowledgeable users are not impressed, and unsophisticated users are still unfamiliar with it. Delta airlines reported that IBM bid Easel on a major program outsourced to IBM, but reversed course a month into the implementation because IBM foresaw implementation limitations and potential performance bottlenecks in Easel's interpreted pseudo-code.
3. Corporate MIS has figured out that SAA is all fluff and no substance. This does not seem to have any adverse consequences towards IBM.
4. IBM is still two years away from demonstrating true cooperative transaction processing using OS/2 Extended Edition (EE) clients with CICS/DB2 mainframe applications.
5. We believe IBM has no choice but to deliver the PC-based client capabilities promised with OS/2 EE as part of SAA, in spite of the industry conclusion that OS/2 is dead.

Conclusions

IBM continues to market mediocre products superbly.

We believe that IBM is, yet again, likely to cause a market backlash because the carefully crafted IBM frontware-based demonstrations are far beyond the capacity of most organizations to implement on their applications today. This places Apple in the awkward position of having to expose the IBM emperor with no clothes.

Until IBM delivers the transaction integrity features which support networked client-server transaction processing with ensured recoverability, the market for mainframe client-server cannot mature. Decision support client-server applications should be the primary focus through 1993, with a secondary focus on the capability to support development of production transaction processing using the client-server model.

Implications

Transaction processing represents an over-\$30 billion market for hardware and software. Most 3270 seats today are used for production transaction processing applications. These seats cannot be easily replaced, except with frontware-based applications, until IBM solves its SAA communications and data integrity problems circa 1993.

APPLE COMPUTER AND DIGITAL EQUIPMENT

Findings

1. Aberdeen visited a large VAXcluster site (Georgia Department of Transportation) to investigate the special relationship, if any, between joint Apple and Digital Equipment sites. Georgia DOT is primarily an engineering-oriented site. They have few Macintoshes because they believe Macintosh is not a credible general-purpose engineering workstation. Thus, our direct research for this project does not support the theory of a special relationship.
2. Other Aberdeen research suggests Digital sites have a bias against IBM products, believing Digital's VAX/VMS is superior to IBM's midrange and mainframe offerings. Digital sites often "march to a different drummer". We believe this bias may extend to PCs, but would require additional research.
3. There are a number of connectivity products which integrate VAX and Macintosh today.

Implications

If Digital VAX sites are biased against IBM PCs, then marketing Apple PCs to VAX sites may have a higher than average success rate.

APPLE COMPUTER AND SOFTWARE AG

Findings

1. Software AG (SAG) has a client software product called Natural Connection which operates on both IBM PCs and Macintosh in conjunction with SAG's Adabas DBMS and Natural 4GL products.
2. Two survey sites (Delta Airlines and University of Texas) were users of Natural. The university also extensively used Natural Connection on Macintoshes. Delta is about to add Macintosh to the list of MIS-approved workstations.
3. Aberdeen's experience is that Natural is highly regarded by its users and our research shows that Natural Connection is one of the most advanced client CASE tools on the market today.
4. Aberdeen believes client CASE is absolutely necessary to speed market growth -- the C-based 3GL programming alternative will take too long and cost too much for MIS to afford.

Conclusions

We recommend an evaluation by Apple of the potential for expanded joint marketing with SAG of the Natural Connection for Macintosh products.

Implications

As MIS increasingly embraces CASE tools for application

development, Apple's ability to run de facto industry CASE tools such as IEW and Excelerator will become a greater stumbling block to MIS sales.

III. APPLICATIONS OF FRONTWARE

This section examines a number of real and hypothetical business applications which use PC-based frontware as a non-invasive front-end to mainframe applications. Aberdeen breaks these applications into three distinct classes: Coordinator, Data Interchange, and Enhanced Display.

Coordinator

The Coordinator class uses frontware to gather information from multiple applications or remote computers. The benefits to the users of Coordinator frontware are:

- O Single login, single request transaction -- no more logging in to multiple applications sessions, each with unique login and data query transactions;
- O Data consolidation -- no more writing down or printing information from one screen before moving on to the next screen to get all the required data;
- O Faster answers and higher productivity -- because data consolidation and single login eliminates human computer "busywork" overhead;
- O Potentially lower mainframe host utilization -- because the frontware can consolidate data quickly and release host resources faster.

Coordinator-class applications are useful to two classes of computer users, casual users and knowledge workers.

Delta Airlines implemented a Coordinator-class application to assist its flight crews, who are casual computer users. The flight schedule, crew assignment and weather data are stored on two different Delta machines. One uses the cryptic PARS transaction coding, while the other machine uses standard CICS conventions. Delta found pilots were frequently using the wrong login protocol for the machines they were accessing, resulting in lost pilot productivity and intense frustration. The Delta frontware system automatically establishes simultaneous sessions on the two different computers, and uses a GUI to speed pilot data selection and display.

Merrill Lynch has numerous applications accessed by remote branch offices.

Because each application may run on a different computer, it is now necessary to dedicate terminals to specific applications such as cash management. There is considerable duplication of coaxial cables and data links. A frontware application will consolidate applications so that brokers, branch managers, and back office applications will all use a common front-end to a branch LAN-based server. The server will have the gateway connections to the remote application host computers.

Apple Computer's SAMS application is a classic example of a Coordinator-class application for knowledge workers. We recommend highlighting Apple's success in front-ending SAMS.

Data Interchange

We define Data Interchange as frontware which translates or transforms mainframe data into a useful display form and optionally passes it to another PC application for additional processing or storage. Data interchange is most effective when used with the color GUI of most PCs.

The benefits to users of Data Interchange are: o Easier comprehension due to color and GUI; o

Ability to transparently use mainframe data in PC applications and vice versa; o Intelligent data storage by the PC of mainframe data, promoting local databases.

A stock trader workstation receives information feeds from numerous sources. The data is typically price quotes or news tickers (e.g. Dow Jones or Reuters news wires). Profitable stock trading depends on reacting to market information before the competition -- literally in seconds. By executing local trend algorithms, graphing price quotes, highlighting significant changes in color, and sounding an alarm on a triggered event, a trader's sensory bandwidth can be enhanced.

The U.S. Navy is implementing a paperless human resource information for job postings using Easel on IBM PS/2's with touch screens. The casual users whose use the system are unaware of the mainframe host as the source of the data.

Executive Information Systems are specialized decision support systems. EIS systems combine Coordinator-class and Data Interchange-class frontware to both hide the host data gathering and to make the information content high, primarily by sorting, summarizing, graphing, and display-by-exception.

Office Information Systems use data interchange with PC-based word processing, desktop publishing, database, and spreadsheet applications in conjunction with e-mail across host-based enterprise mail networks. IBM's OfficeVision/2 and Apple's MacPROFS are two examples of such office information systems.

Enhanced Display

Enhanced Display applications improve the display content of host data, but add little if any additional value. The so-called Pretty Face applications use frontware to add color and GUI-based symbols like radio buttons and sliders to replace 3270 function keys.

Enhanced Display applications are easiest to implement but essentially add no more value than mere terminal emulation.

IBM is demonstrating enhanced display frontends to many of its mainframe utilities, conditioning the market to enhanced display frontware. However, more sophisticated frontware such as OfficeVision/2 which adds significant value to data at the PC is being routinely delivered late.

Aberdeen is concerned that the market will begin to lump all frontware into the Enhanced Display class, slowing acceptance of the truly useful Coordinator and Data Interchange classes of frontware applications.

IV. INTERVIEW CONTACT SUMMARIES

Aberdeen Group conducted in-person interviews with the following organizations and persons during the course of the study. Interviews were granted with the provision they not be used for sales or promotional purposes, and that the respondent's names and organizational identity not be published.

Organization	City	Using
Atlanta Gas & Light	Atlanta, GA	IBM 3090
Continuum Co	Austin, TX	IBM 3090
Delta Airlines	Atlanta, GA	IBM 3090s
Federal Reserve Bank of NY	New York	IBM 3090s
First Interstate Bank, Ltd	Los Angeles	IBM 3090s, Stratus
Georgia Dept of Transportation	Atlanta, GA	DEC VAX/VMS
Merrill-Lynch	New York	IBM 3090s
Prudential-Bache	New York	IBM 3090s Sabre Group
IBM 3090s		Atlanta, GA
United Service Auto Assn	San Antonio, TX	IBM 3090s
University of Texas	Austin, TX	IBM 3090

Our interview sites were selected from lists of mainframe computer sites where we could identify client-server computing decision makers. Geographic diversity was achieved by conducting interviews in Atlanta, Los Angeles, New York, and Texas.

Methodology

In-person interviews were conducted in three phases:

- O Information about the enterprise, computing environment, terminal environment, corporate PC policies, budgets, and interviewee's resume. During this phase, the respondent was unaware that Apple was sponsoring the research;
 - O A ten-minute presentation was made using the slides and script included in Appendix A. The respondent was told about Apple and MitemView;
 - O The respondent was asked for subjective opinions about cooperative processing, Apple Computer, frontware, and their organization's likely path towards cooperative processing.
- Aberdeen Front End Survey -- Atlanta Gas Light

Aberdeen Group and Apple Computer Confidential
Interviewer Peter Kastner

Page

1. Enterprise Profile

Name/Division Atlanta Gas Light Company

Location

1219 Caroline Street
Atlanta, GA 30302

Type of business Atlanta-area public utility

(Telephone) (404) 584-4136

(Size of company) 3,800 employees

(Major competitors) None

2. Respondent Profile

Name Carlos Tavares Director, New Applications

Address

Telephone

Responsibilities

Any new systems. Rewriting mission-critical customer utility service application.

Background and experiences: Ten years here.

Experience with IBM and Apple PCs: IBM PCs for 6 months.

3. Computing Environment

Mainframe Configuration

3090-400 MVS/XA going to ESA. Ideal DBMS moving to DB2.

Midrange & Distributed Systems

None

Key Applications

1.Utility customer service - CIS. Seventy Andersen Consulting programmers are rewriting it over 2 1/2 years.

4. Terminal Environment

Mainframe

3270 terminals

3270 active users (local & remote)

2,000+ across state

PC LAN users

Which key applications use 3270's

Local PCs & LAN's without mainframe connectivity Major switch to PCs over the next four years.

5. Corporate PC Policy

Who approves PC Purchases Each department

Who approves PC suppliers Purchasing

Approved PC manufacturer list IBM, Compaq

Who sells and installs the PC's Inacomp

6. Apple Computer Products Installed

Number of Macintoshes installed A few. None in MIS.

Macintosh percentage of total PCs Trivial

What departments/applications use Macintoshes

1991 buying plans None

7. Using Personal Computers in production applications

Applications which today use the client/server model
None

Plans for new applications to use the client/server model
No

What client platforms are you considering

Evaluation of client software environments (subjective comments) None -- does not feel confident in product knowledge.

Microsoft Windows 3.0

Hewlett-Packard NewWave

IBM OS/2 Presentation Manager

SCO Open Desktop

Sun Open Look

Apple Computer Macintosh

Other

How will the MIS organization develop expertise in client/server systems development

Enterprise commitment to IBM's SAA

Not really

8. Budgets and Sizing Estimates

1990 Systems and Programming Budget \$ (excludes ops)

1991 S&P budget growth \$ or %

Flat

1990 Head Count
house

40 in S&P + 70 out-of-

Applications Budget Breakdown (total 100% -- indicate people or \$)

50%	- Maintaining/enhancing existing applications	
	- Designing/programming new applications	25%
	- User training and education	5%
	- Systems management and Other	20%

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions?

Don't know much about Apple. Great graphics. Good company.

Were you aware Apple had such capabilities? No.

Have you considered a front end to existing applications such as the MitemView product I described?

No, we are an IBM shop. Over the past year we have moved closer to IBM. We dropped Telex for 3270's for instance.

Would a front end solve a real problem for you? (what, why)

Probably not. We are having more than enough problems dealing with front-end CASE.

Which are you likely to invest in

- leave existing applications alone Yes

- front end to existing application such as MitemView
No

- develop client server model cooperative processing
Don't know

How much over what time frame
Now and 1993

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh? Don't know.

What do you think of Apple's recent changes in product price/performance No notice.
Occasionally read ComputerWorld or Byte.

Are dealers able to meet your PC/LAN support requirements
Yes

What should Apple do to improve its sales to enterprises like yours [laugh] Convince
executive management that IBM can't do the job!

12. Other notes

A.) Using Knowledgeware for CASE. Having problems.

B.) IBM has visible presence here. Tech support group focus. No Apple presence.

C.) Company says it is an engineering company. Engineers dictate software MIS
will install & support.

D.) CASE - no in-house methodology yet selected. Playing. Failure in the making.
No direction. "The platform is not there." Eight trained, two left. "MIS has a bad reputation
inside Atlanta Gas. CASE is going to make our reputation worse."

E.) Wants Apple to run same CASE tools as generally available on IBM PC.

Interviewer's Comments: A trailing edge site in a conservative industry. Likely to follow IBM
anywhere. Unable to understand the content of a technical sell.

Name/Division Continuum Company

Location Austin, Texas

Type of business Software Developer - Insurance Systems.
Continuum was chosen as a representative of the
mainframe application developers and remarketers.

(Telephone) (512) 345-5700

(Size of company) \$62M in FY89; \$80M in FY90

(Major competitors) EDS, Policy Management Systems, Cybernetics

2. Respondent Profile

Name Ron Rohrbach Manager, Workstation Development

Address 9500 Arboretum Blvd., Austin, TX 78759-6399

Telephone (512) 338-7460

Responsibilities

- 1.) Supporting worldwide customer workstation requirements
- 2.) Strategy & planning for company's client-server products

Background and experiences:

3 years here. Extensive implementation background at Ernst & Whinney,
and HP.

(Experience with IBM and Apple PCs)

3 years in-depth with PCs. Boss has both PC & Mac. Expects to have a Sun
workstation by next year. Very opinionated.

3. Computer Environment

Mainframe Configuration

Company's 400 customers almost all use IBM MVS. Main product is CCA, an IMS/DB2
system for insurance policy management.

Midrange & Distributed Systems None
supported.

Key Applications

- 1. Life insurance systems

4. Terminal Environment

Mainframe	"We support anything IBM supports"
3270 terminals	mostly
3270 active users (local & remote)	
PC LAN users	using 3270 emulation - growing
Which key applications use 3270's	All applications

Local PCs & LAN's without mainframe connectivity

5. Corporate PC Policy Not Applicable

Who approves PC Purchases

Who approves PC suppliers

Approved PC manufacturer list

Who sells and installs the PC's

6. Apple Computer Products Installed

Number of Macintoshes installed	35 for in-house use
Macintosh percentage of total PCs	4%

What departments/applications use Macintoshes

1991 buying plans

- units/models
- change in Macintosh/PC mix percentage
- current attitude towards Apple niche player

7. Using Personal Computers in production applications

Applications which today use the client/server model

None. But it's on the lips of every IBM sales rep and our customers are listening. We are asking whether the concept is industrial strength today? Despite the over-hype, we will support the philosophy of client-server.

Plans for new applications to use the client/server model

Over the next several years. Plans are presently proprietary.

What client platforms are you considering

- names IBM
- reason Our Customers buy IBM

Evaluation of client software environments (subjective comments)

Microsoft Windows 3.0 -- Superior product. Next year it will hide DOS

Hewlett-Packard NewWave -- Good reviews. No experience. Could be a winner but HP can't market anything.

IBM OS/2 Presentation Manager -- Only Extended Edition is worth talking about.
Users beware - too sophisticated &

expensive.

SCO Open Desktop -- Unix is confusing so users are confused.

Sun Open Look -- Motif wins

Apple Computer -- Niche player. Maybe recent lower price will change attitudes, but I'll wait & see.

Other

How will the MIS organization develop expertise in client/server systems development We will train ourselves

Enterprise commitment to IBM's SAA -- SAA is an idea whose time has already past. Any products now are too late. I have to develop my own equivalents.

8. Budgets and Sizing Estimates Not Applicable

1990 Systems and Programming Budget \$ (excludes ops) \$

1991 S&P budget growth \$ or %

1990 Head Count

- Analysts
- Applications programmers
- Systems programmers, DBAs, network mgr
- PC infocenter
- User training and education

Applications Budget Breakdown (total 100% -- indicate people or \$)

- %
- Maintaining/enhancing existing applications
 - Designing/programming new applications %
 - User training and education %
 - Systems management and Other %

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions? Apple is just not the first company I think of for client workstations. Their products have been overpriced and, more importantly, there is much more software tools available for PC than the Macintosh.

Were you aware Apple had such capabilities? No. How can Apple expect to sell if they don't get out on the street? Wants more info on Mitem and Apple client products.

Have you considered a front end to existing applications such as the MitemView product I described?

No -- we plan to go directly to client-server cooperative processing.

Would a front end solve a real problem for you? (what, why) Already VAR Easel
& In-front plus two he can't disclose.

Which are you likely to invest in

- leave existing applications alone No.
- front end to existing application such as MitemViewSome, probably PC-based.
- develop client server model cooperative processingYes.

How much over what time frame 3 years

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh
They caught up. For 90% of users, MS Windows 3.0 equals a Mac.

What do you think of Apple's recent changes in product price/performance Helps, may be too late.

Are dealers able to meet your PC/LAN support requirements

Yes. We don't need dealer support -- we work directly with hardware and software suppliers.

What should Apple do to improve its sales to enterprises like yours

- 1.) Articulate a business reason for Macintosh over PC
- 2.) Call on us. Talk to us. I didn't know about the Mac workstation, Mitem, and other products discussed
- 3.) IBM provides free support and lots of hand-holding. My experience is that Apple charges for everything -- if they even offer it.

12. Other notes

A.) No one can afford to keep developing clients using C and the Windows toolkit. We'll use high level languages and productivity enhancers. B.) Thinks Geo DOS may be a GUI winner on older PCs.

C.) The insurance industry follows IBM's lead and buys IBM, not substitutes.

D.) Impression is Mac is expensive and therefore not useful as a commodity workstation.

E.) Apple hasn't been able to get a sales foot in the MIS door. They aren't considered a player due to a total lack of feet-on-the-street marketing.

13. Interviewer's Notes

Aberdeen also spoke with Dun & Bradstreet Software about their plans for client-server applications and frontware. D&B already has begun delivering the ViewSeries of frontware for the former MSA and M&D accounting packages. D&B has no plans, based on market requirements, for offering the ViewSeries on Apple Macintosh.

1. Enterprise Profile

Name/Division	Delta Airlines
Location	Hartsfield Airport & Virginia Avenue Atlanta, GA 30320
Type of business	International airline
(Telephone)	(404) 765-4055
(Size of company)	\$8.1 billion
(Major competitors)	

2. Respondent Profile

Name	Mark F. Addison	Supervisor, Information Center
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Address

Telephone

Degree of difficulty in reaching the right respondent Moderate

Responsibilities

Emerging technology evaluation: client-server, imaging, workstations.

Background and experiences:

Here 10 years. 4 years managing the Infocenter.

Experience with IBM and Apple PCs:

Has 2 Macs & 2 PCs in his office. A hands-on user who works closely with staff on product evaluations.

3. Computing Environment

Mainframe Configuration

15-20 3090's. 8 running TPF; MVS, VM. Using SAG Adabas database 5 years. One DB2 application.

Midrange & Distributed Systems

10-15 VAXes. Audio response. Maintenance & scheduling. Lead tracking. Many systems were outsourced.

Key Applications

- 1.Reservations
- 2.OSS - operations support
- 3.TOS - tech operations

Most applications run under a TPF umbrella.

4. Terminal EnvironmentMainframe

6-7,000	3270 terminals	Westinghouse Reservations terminals:
	3270 active users (local & remote)	< 50
	PC LAN users	4,500 on Novell LANS
	Which key applications use 3270's	None
	Local PCs & LAN's without mainframe connectivity	
	Few and dropping	

5. Corporate PC Policy

Who approves PC Purchases Weekly Management Committee review

Who approves PC suppliers	Infocenter
Approved PC manufacturer list	IBM 55SX
Who sells and installs the PC's	Maintained in-house

6. Apple Computer Products Installed

Number of Macintoshes installed	A few
Macintosh percentage of total PCs	Very low
What departments/applications use Macintoshes	
1991 buying plans	
- units/models	Several hundred
- change in Macintosh/PC mix percentage	Towards more Macs
- current attitude towards Apple	Delta will probably approve
Mac for general use as a PC within 90 days.	

7. Using Personal Computers in production applications

Applications which today use the client/server model
None

Plans for new applications to use the client/server model Employee Insurance,
HR, flight control

What client platforms are you considering

- names IBM PC 55SX

- reason Approved PC

Evaluation of client software environments (subjective comments)

Microsoft Windows 3.0

Employee group application pilot

Hewlett-Packard NewWave Like it but tied to HP servers. A heavy Novell user. Real benefit is servers.

IBM OS/2 Presentation Manager One flight control application going into production. Needed multithreading capability of OS/2.

SCO Open Desktop

None

Sun Open Look Some in network monitoring. Evaluating IBM RS/6000

Apple Computer Macintosh Approve soon for general use.

Other

How will the MIS organization develop expertise in client/server systems development

Flight control application is a learning tool. In-house development and training now being planned.

Enterprise commitment to IBM's SAA

Will definitely use CUA. Will consider SAA client-server.

8. Budgets and Sizing Estimates

1990 Systems and Programming Budget \$ (excludes ops)
Over \$100M

1991 S&P budget growth \$ or % approval basis

Not available, Project

1990 Head Count

About 1,800

- Analysts

- Applications programmers

- Systems programmers, DBAs, network mgr

- PC infocenter

- User training and education

Applications Budget Breakdown (total 100% -- indicate people or \$) 75 people in the Infocenter. About 20 are in development.

45%	- Maintaining/enhancing existing applications	
	- Designing/programming new applications	25%
	- User training and education	15%
	- Systems management and Other	15%

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions?

Pleased with Apple over a one-year sales evaluation period. Apple did a superior selling job throughout all levels of the Delta buying chain.

Were you aware Apple had such capabilities? Vaguely. Not aware of Mitemview.

Have you considered a front end to existing applications such as the MitemView product I described?

Yes. Did it with Automator M1 (UK) for a flight control application. Better than what we had. Front-end hid the vastly different user terminal styles between TPF and CICS.

Would a front end solve a real problem for you? (what, why)

Yes. Will solve employee insurance application. Throwing out a Wang VS for PCs on a Novell LAN.

Which are you likely to invest in

- | | | | |
|-----|---|---|------|
| | - | leave existing applications alone | Some |
| Yes | - | front end to existing application such as MitemView | |
| | - | develop client server model cooperative processing | Yes |

How much over what time frame

It will be 2-3 years before it becomes commonplace at Delta.

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh?

Apple still has a usability lead, but there are features in Windows and NewWave which Apple should adopt.

What do you think of Apple's recent changes in product price/performance
Aware of the new products. Delta will rely heavily on Apple to configure initial systems because there is not enough in-house experience at rule-of-thumb configuration.

Are dealers able to meet your PC/LAN support requirements
Delta services PCs in-house. Selecting Apple as a PC here has large implications for Infocenter and maintenance logistics.

What should Apple do to improve its sales to enterprises like yours
We got started a year ago looking at Apple because a VP thought Apple might be able to do the job. Apple direct sales has done an excellent job selling and supporting us during the evaluation period.

12. Other notes

A.) Challenge: there is no application development cookie-cutter. Everything is custom. LAN management driving us crazy in operations, administration. We are about to get Personnel approval to add one-half person in LAN management headcount for every sixteen people added to LANs.

B.) CASE has not yet penetrated Infocenter. Software AG Natural is increasingly used for host applications.

C.) New scheduling & Aircraft Tech Support applications are now using Adabas Natural.

D.) Easel: IBM dropped Easel for CASE/PM on flight management application
due to interpretive nature of run time and the likely performance impact.

E.) Anderson Consulting has 200 people at the Delta jetbase writing PC bar-code/APPCC/DB2 applications.

F.) No apparent selling by Software AG of Natural Connection to the Infocenter.

Aberdeen Group and Apple Computer Confidential Page
Aberdeen Front End Survey -- Federal Reserve Bank of New York
Interviewer Peter Kastner

1. Enterprise Profile

Name/Division Federal Reserve Bank of New York

Location 59 Maiden Lane

New York, NY

Type of business Quasi - government financial institution

(Telephone) (212) 720-5767

(Size of company)

(Major competitors)

2. Respondent Profile

Name Ms. Vera Crout Manager, Advanced Technology

Responsibilities

Recently made manager of front-line group to study use of advanced technology: CASE, imaging, voice, expert systems, EIS, distributed computing including cooperative processing.

Background and experiences:

Managed distributed systems group

(Experience with IBM and Apple PCs)

User of office system. Has ancient Apple Lisa and a PC on her desk.

3. Computer Environment

Mainframe Configuration

IBM 3083 + 3090 + Hitachi. Running MVS.

Oracle to DB2 distributed database conversions underway

Midrange & Distributed Systems

DG Eclipses: PC file server, departmental computing, office automation using CEO. Oracle is the DBMS. SAS for analysis.

Key Applications

1.Funds transfer -- largest FRB FedWire

2.Securities

3.Open Market activities

4. Terminal Environment

Mainframe

3270 terminals

Was 700. Now 100.

3270 active users (local & remote)

PC LAN users

1,100

Which key applications use 3270's

PC is now the defacto workstation. Applications have evolved from 3270 to 3270 Irma board to LAN - connected to DG.

Local PCs & LAN's without mainframe connectivity About 100 will remain standalone.

5. Corporate PC Policy

Who approves PC Purchases Capital budget process. Infocenter support.

Who approves PC suppliers

Approved PC manufacturer list 286 AT or 386
IBM or DG

Who sells and installs the PC's 6.
Apple Computer Products Installed

Number of Macintoshes installed A couple.
Macintosh percentage of total PCs

What departments/applications use Macintoshes

1991 buying plans

- units/models Nothing significant.
- change in Macintosh/PC mix percentage
- current attitude towards Apple Not part of our strategic plans.

7. Using Personal Computers in production applications

Applications which today use the client/server model None.

-application description
FedLine is a PC-based "on-line batch" service for member banks. Had one old Burroughs client-server application.

Plans for new applications to use the client/server model
Have established client-server laboratory in advance of new applications.

What client platforms are you considering

- names Will evaluate and decide in 1991.

Evaluation of client software environments (subjective comments)

Microsoft Windows 3.0

Will evaluate

Hewlett-Packard NewWave Using in an experiment. Looks like a fantastic product. CEO office uses it.

IBM OS/2 Presentation Manager Will evaluate, but nobody here is demanding it.

SCO Open Desktop

Will evaluate it

Sun Open Look

2 Sun's in research. DG Aviion

Apple Computer Macintosh No

Other

Knowledgeware IEW for CASE

development is the FRB standard. First project is in a pilot.

How will the MIS organization develop expertise in client/server systems development
Study in laboratory. Evolve standards.

Enterprise commitment to IBM's SAA

Committed. The PS/2 is the SAA desktop platform. We are an IBM PC-compatible shop.

8. Budgets and Sizing Estimates

1991 S&P budget growth \$ or %

Up but not a high-growth year.

1990 Head Count

- Analysts 220 in S & P
- Applications programmers
- Systems programmers, DBAs, network mgr
- PC infocenter
- User training and education

Applications Budget Breakdown (total 100% -- indicate people or \$)

- 60% - Maintaining/enhancing existing applications
- Designing/programming new applications 30%*

- User training and Systems management 10%
- * Eliminated Burroughs which caused high IBM development rate

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions? Not interested.

Were you aware Apple had such capabilities? No

Have you considered a front end to existing applications such as the MitemView product I described?

Why front end when you can get the full benefits of cooperative processing?

Would a front end solve a real problem for you? (what, why) Only for applications which will not be changed for a long time.

Which are you likely to invest in

- leave existing applications alone
- front end to existing application such as MitemView
- develop client server model cooperative processing Yes

How much over what time frame

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh Nothing is unique about the Apple interface anymore.

What do you think of Apple's recent changes in product price/performance Still more expensive than a PC.

Are dealers able to meet your PC/LAN support requirements

What should Apple do to improve its sales to enterprises like yours

The whole financial industry bought the IBM PC. Now, there is an overwhelming amount of software for the PC. Where are the applications for Apple?

12. Other notes

A.) At the time the FRB chose the IBM PC as a workstation, Apple could not support the RFP functional requirements. The Bank made a long-term commitment to the IBM PC architecture. Apple then was a closed box with limited expandability -- and pricey as well.

B.) Apple should license Sun Sparc and be a name-brand open systems company.

C.) Client-server benefits will only come with corresponding operational workflow changes.

Aberdeen Group and Apple Computer Confidential	Page
Aberdeen Front End Survey -- First Interstate Bank, Limited	
Interviewer Peter Kastner	

1. Enterprise Profile

Name/Division	First Interstate Bank, Limited
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Location	707 Wilshire Blvd. Los Angeles, CA 90017
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Type of business	Large regional and international bank
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(Telephone)	(213) 614-3616
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(Size of company)	
-------------------	--

(Other enterprise data)	The Limited Bank is responsible for all commercial banking activities as part of First Interstate Bank
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2. Respondent Profile

Name	John J. Beale Vice President and Manager Also present: Terry Stallcop, V.P. Dennis Ceru, V.P.
------	--

Responsibilities	Cash management and money transfer systems, future architecture and technology planning for the Limited Bank.
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Background and experiences:

Ten years in banking systems here and at Citibank.

Experience with IBM and Apple PCs:

Limited

3. Computing Environment

Mainframe Configuration

3090 - 600's and Stratus XA 2000

Midrange & Distributed Systems

Key Applications

1.Cash Management

2.Lock Box

3.Funds Transfer

4. Terminal Environment

Mainframe

3270 terminals

Fewer than 100

3270 active users (local & remote)

PC LAN users

A few. (Branch has LANS for tellers)

Which key applications use 3270's

Local PCs & LAN's without mainframe connectivity

Most -- about 6,000 PCs

5. Corporate PC Policy

Who approves PC Purchases DP Policy > \$100K

1) Graphics

Who approves PC suppliers

No list

2) Deft CASE

Approved PC manufacturer list

No list

3) Design model

Who sells and installs the PC's

Dealer

Note: Even the FIB ad/

slide dept doesn't use Macs

6. Apple Computer Products Installed

Number of Macintoshes installed

20

Macintosh percentage of total PCs 1%

Toy, nonstandard, not business tool, needs more applications, sheer numbers of PCs at our own and customer sites.

What departments/applications use Macintoshes

Beale's department has less than 5 Macs, predominantly used for the Deft CASE product, a design front-end to Sybase SQL Server.

1991 buying plans

- units/models none

- change in Macintosh/PC mix percentage none

- current attitude towards Apple

"Mac is a toy", "it's not a business tool", "needs more applications", "of no interest...all our customers use IBM PCs, so we support IBM PCs"

7. Using Personal Computers in production applications

Applications which today use the client/server model
None

Plans for new applications to use the client/server model

Electronic PC Window, a customer client based on the IBM PC, will be in operation by the end of 1991.

What client platforms are you considering

- names IBM PC

- reason Claybanc Systems (Oregon) & Stockholder Systems Inc. (Atlanta) have cash management client software.

Evaluation of client software environments (subjective comments)

Microsoft Windows 3.0 Considering Hewlett-Packard NewWave No interest

IBM OS/2 Presentation Manager Considering

SCO Open Desktop No interest

Sun Open Look Perhaps for future servers

Apple Computer Macintosh No interest

Other

How will the MIS organization develop expertise in client/server systems development
 Corporate: outsource
 Limited Bank: internal and external

Enterprise commitment to IBM's SAA
 Weak and decreasing. "There's nothing there!"

8. Budgets and Sizing Estimates

1990 Systems and Programming Budget \$ (excludes ops)
 \$ 7-9M includes \$2M for development
 1991 S&P budget growth \$ or % in savings over 2 years. Flat. Need to find \$1M

1990 Head Count 80 people

- Analysts
- Applications programmers
- Systems programmers, DBAs, network mgr
- PC infocenter
- User training and education

Applications Budget Breakdown (total 100% -- indicate people or \$) + Capital

\$1.7M

Costs

- Maintaining/enhancing existing applications \$1,500K
- Designing/programming new applications 550K
- User training and education 50K
- Systems management and Other 50K

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions?

Mac must become IBM PC compatible. Not considered because: Must do Lotus 123 or Symphony as it's a Cash Management institutional standard. Lack of salability. Price competitively. Same PC comm: Ethernet + Token Ring needed.

Were you aware Apple had such capabilities?

Nothing the bank uses is mouse-based -- all character-based today. Other than the stock trading floor, no bank use for GUI applications.

Key business model: don't make customer change the way they work. Mac equals change. Customer does not want to change work habits. But dichotomy is application customization is needed to meet high service levels.

Bank does believe in cooperative processing based on costs, price performance, and quicker development.

Which are you likely to invest in

- leave existing applications alone
- front end to existing application such as MitemView

No

- develop client server model cooperative processing
- Yes -- project already underway

How much over what time frame Deliver starting at the end of 1991.

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh?

Yes

What do you think of Apple's recent changes in product price/performance

Price is not why people pass by Apple; the IBM PC is the corporate workstation standard. However, a cheap PC is much cheaper than a Mac.

Are dealers able to meet your PC/LAN support requirements

Yes

What should Apple do to improve its sales to enterprises like yours

Get our customers, who are large-company corporate cash managers and treasurers, to demand Apple. We will respond to market demand, but right now there is absolutely none for Apple.

12. Other notes

A.) Apple needs customization and low cost, IBM compatibility, and efficiency

B.) Bank market research: there are no application users today for Mac cash management.

C.) Perception: Mac is a graphics machine. Never marketed to corporate America.

D.) Division's Ethernet/Token Ring PC LAN is not AppleTalk compatible.

Aberdeen Group and Apple Computer Confidential Page
Aberdeen Front End Survey -- Georgia Dept. of Transportation
Interviewer Peter Kastner

1. Enterprise Profile

Name/Division Georgia Department of Transportation

Location Two Capital Square, Room 183
Atlanta, GA 30334

Type of business State Agency

(Telephone) (404) 656-6034

(Size of company)

(Major competitors)

2. Respondent Profile

Name Ben Maxwell Manager, Systems & Programming

Address

Telephone

Responsibilities

Applications Support. Eighty people.

Background and experiences:

Civil engineer by training.

Experience with IBM and Apple PCs:

Just a little general PC knowledge. Considerable engineering workstation experience.

3. Computing Environment

Mainframe Configuration the
VAX

Midrange & Distributed Systems

VAXCluster 2X6330, 8700, 8350. Adding MV 3800 and MV 4000 in
LAVC

Key Applications

- 1.Engineering-based applications
- 2.Road design (CCMC and IRDS)
- 3.Project planning and scheduling
- 4.Traffic & accident

Accounting is via State IBM complex -- separate

4. Terminal Environment

DEC VT/PC mix depends on a person's job requirements.

Mainframe

3270 terminals

A few. Switching to VAX SNA gateway.

3270 active users (local & remote)

PC LAN users

225 + 90 design PC's - 12 VAX

5000's. Everything goes DOT Ethernet. Everybody happy with bandwidth. Which key applications use 3270's 75-80 are graphics.

DEC VT terminals count is 400. Slowly replacing with VT 320/420. Used for data entry, All-in-One.

Local PCs & LAN's without mainframe connectivity

Intergraph Microstations with CCMC software. Not actively pursuing geographic information systems (GIS) but might change -- big impact on computing requirements.

5. Corporate PC Policy

Who approves PC Purchases Capital budget item

Who approves PC suppliers

Maxwell

Approved PC manufacturer list

IBM & Compaq. IBM for design

Who sells and installs the PC's

State purchasing contract

6. Apple Computer Products Installed

Number of Macintoshes installed

A handful in traffic department for accident reporting. Key reason is accident scene diagrams.

Macintosh percentage of total PCs Very low

What departments/applications use Macintoshes

1991 buying plans None

- units/models
- change in Macintosh/PC mix percentage
- current attitude towards Apple

7. Using Personal Computers in production applications

Applications which today use the client/server model None

Plans for new applications to use the client/server model

By late 1991, will look to department-wide DB and client/server. Power & low cost compelling us to take advantage of it. Less training required.

What client platforms are you considering

- names Digital PCSA
- reason Digital's PC support product

Evaluation of client software environments (subjective comments)

None to date -- no comments

Microsoft Windows 3.0

Hewlett-Packard NewWave

IBM OS/2 Presentation Manager

SCO Open Desktop

Sun Open Look

Apple Computer Macintosh

Other DEC Windows on Workstation. Use PCSA on PC's.

How will the MIS organization develop expertise in client/server systems development
In-house. Unsure how fast they might move to enterprise database.

Enterprise commitment to IBM's SAA

None

8. Budgets and Sizing Estimates

1991 S&P budget growth \$ or % cut in FY92. Flat with 1990. May

1990 Head Count 80 (15 data entry)

- Analysts
- Applications programmers
- Systems programmers, DBAs, network mgr
- PC infocenter
- User training and education

Applications Budget Breakdown (total 100% -- indicate people or \$)

60%

- Maintaining/enhancing existing applications
- Designing/programming new applications 20%
- User training and education 10%
- Systems management and Other 10%

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions?

Bought the PC's for engineering design. Would require Apple to run their CAD software. Large application selection needed. At major sites: WP, DB, spreadsheet, VAX connectivity, dBase III Plus, Lotus 1-2-3.

Were you aware Apple had such capabilities?

No

Have you considered a front end to existing applications such as the MitemView product I described?

No

Would a front end solve a real problem for you? (what, why)

No -- no need with engineering packages. Perhaps some need when VAX DBMS ties to CAD in 1992.

Which are you likely to invest in

- leave existing applications alone Probably

No - front end to existing application such as MitemView

Slowly - develop client server model cooperative processing

How much over what time frame
1992 local and remote file server in FY92.

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh?
Yes. His impression caused by Windows 3.0 ad campaign.

What do you think of Apple's recent changes in product price/performance Saw ads. Got message. Didn't apply to me.

Are dealers able to meet your PC/LAN support requirements Yes. State contract.

What should Apple do to improve its sales to enterprises like yours
Must concentrate effort on technical market capabilities & compatibilities with existing applications (shrink-wrapped or US-wide DOT). Apple has not made a real effort to market to us. Price/performance not the only consideration. Suggests ads and trade shows aimed at highway engineers.

12. Other notes

A.) Apple has innovative ideas but proprietary SW. Neutral on Apple as a company. Haven't seen compelling reason to buy. Apple must get state contract to sell anything to us.

B.) Impression is Mac not used for engineering, technical, and design.

C.) Would consider extended trial.

D.) Are just purchasing SCO Unix for evaluation.

E.) DEC - good account team. Long term relationship.

Interviewer Peter Kastner

1. Enterprise Profile

Name/Division Merrill-Lynch

Location 225 Liberty Street -- WTC2-9th Floor
New York, NY 10080-6109

Type of business Largest U.S. Brokerage Firm

(Telephone) (212) 236-9170

(Size of company)

(Major competitors)

2. Respondent Profile

Name Thomas Musante
VP & Dept. Manager, Information Strategies Group.
"Risk managers for new technology"

Responsibilities

Picks cooperative processing tools and products. Common data access application interface tools, user interface, code base. Front-end to DB2. Strategy is to reduce programs and programmers.

Background and experiences:

Early thirties. Six years at Drexel Burnham Lambert; A year at DG. Seven years at Equitable Insurance.

(Experience with IBM and Apple PCs)

All DOS machines. No UNIX. Some Windows programming experience.

3. Computer Environment

Mainframe Configuration
IBM 3090s running MVS and VM

Midrange & Distributed Systems

Key Applications

1. Portfolio management
2. Customer accounting and securities handling
3. Branch office management

4. Terminal Environment

Mainframe	
3270 terminals	A few still in clerical activates.
3270 active users (local & remote)	
PC LAN users	15,000 branch PCs + 8,000 for other professionals.
Which key applications use 3270's	
Local PCs & LAN's without mainframe connectivity	Almost none. All 23,000 seats.

5. Corporate PC Policy

Who approves PC Purchases

Who approves PC suppliers IBM & Compaq. Clones are allowed exceptions.

Approved PC manufacturer list	IBM PS/2 & Compaq.
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Who sells and installs the PC's	Dealer -- sometimes IBM
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6. Apple Computer Products InstalledNumber of Macintoshes installed

Macintosh percentage of total PCs

What departments/applications use Macintoshes Sprinkled around but not a DP production platform.

1991 buying plans

- units/models Unknown -- not a DP purchase
- change in Macintosh/PC mix percentage
- current attitude towards Apple

7. Using Personal Computers in production applications

Applications which today use the client/server model No production client-server.

-application description

Plans for new applications to use the client/server model

Many if not most applications. Benefits are at the server side. Two-tier topology following IBM model by executive VP fiat, but data servers at productivity nodes.

What client platforms are you considering

- names 386SX PC. Maybe UNIX boxes. Sun Sparc and RS 6000
a possibility.

- reason

Evaluation of client software environments (subjective comments)

Microsoft Windows 3.0 As end user, I'm delighted. Future
looks good. Some Novell problems. Have established themselves as a standard.

Hewlett-Packard NewWave Architecture is the key. Agents are pre-
emptive.

IBM OS/2 Presentation Manager Late and slow.

SCO Open Desktop Only academic. Looking at OSF DCE. CUA
product not available yet. PM or Win not compliant.

Sun Open Look Yes, because of use of Sun in trader and
analytic departments.

Apple Computer Macintosh Hardly a presence in DP. Mac lacks
connectivity and applications. No demand for it.

Other

How will the MIS organization develop expertise in client/server systems development

Internal

Enterprise commitment to IBM's SAA
Common sense CUA, API. Will use AD/Cycle for development.

8. Budgets and Sizing Estimates

1990 Systems and Programming Budget \$ (excludes ops) \$175M

1991 S&P budget growth \$ or %

1990 Head Count More than 1,900 programmers.

- Analysts

- Applications programmers

- Systems programmers, DBAs, network mgr
- PC infocenter
- User training and education

Applications Budget Breakdown (total 100% -- indicate people or \$)

- 66% - Maintaining/enhancing existing applications
- Designing/programming new applications 33%

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions?

Huge investment here in IBM applications (WP, OA), unamortized PC hardware, network interfaces, service/training knowledge.

Merrill Lynch wants to develop on a single preferred client platform.

Were you aware Apple had such capabilities?

No. Apple has never called on me here or at Drexel.

Have you considered a front end to existing applications such as the MitemView product I described?

Yes. There is a huge dependency on not changing the CICS screens. Requires considerable software distribution issues. Evaluated Easel, Viewpoint & InFront.

Would a front end solve a real problem for you? (what, why)

Have not recommended their use in production systems due to change velocity on mainframe. Also, where is the business benefit? Better to go directly to client-server.

Which are you likely to invest in

- leave existing applications alone
- No

- front end to existing application such as MitemView
- Infrequently

- develop client server model cooperative processing

Preferred migration path. Need to bridge multiple data and processing variations on client-server.

How much over what time frame

A couple of native APPC applications will go into production in 1991. Productivity tools like APPC generators available in 1992. Expects common use in 1994.

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh

I can't think of anything a Mac can do that a PC can't, but I just don't know much about Apple.

What do you think of Apple's recent changes in product price/performance Aware of it.

Are dealers able to meet your PC/LAN support requirements Yes for PCs here in New York.

What should Apple do to improve its sales to enterprises like yours

Hard to convince Merrill management that Apple costs would be lower given the huge commitment and knowledge base for IBM PCs.

12. Other notes

A.) Four client-server models at Merrill-Lynch

- Data Access -- Gupta client to Oracle or Sybase
- Messaging -- APPC
- RPC -- a Netwise user
- Frontware -- Easel, InFront, Mitem

Aberdeen Group and Apple Computer Confidential
Aberdeen Front End Survey -- Prudential Bache
Interviewer Peter Kastner

Page

1. Enterprise Profile

Name/Division Prudential - Bache Securities

Location 199 Water Street - 19th Floor
New York, NY 10292

Type of business Large Brokerage Firm

(Telephone) (212) 214-2131

(Size of company)

(Other enterprise data)

2. Respondent Profile

Name Ms. Loraine Cronin Vice President, Retail Systems

Responsibilities

Manages present and future projects for retail broker automated systems at Bache branch offices.

Background and experiences:

Started in DP developing programmer training. Ran Bache's broker desktop automation project.

Experience with IBM and Apple PCs: Eight years as a user.

3. Computing Environment

Mainframe Configuration

IBM 3090/400. Use 3270 data stream to server at each branch. Download customer data to Convergent Technologies (now Unisys) CTOS-based controller.

Midrange & Distributed Systems

Key Applications

- 1.Client data (local)
- 2.Portfolio Monitoring
- 3.Online ADP quote service
- 4.Spreadsheet and WP

4. Terminal Environment

Mainframe

3270 terminals	3,000
3270 active users (local & remote)	
PC LAN users	8,000 CTOS users; 2,000 Bunker-

Ramo users Which key applications use 3270's

Syndicate system, budgets, exchange, order processing. No plans to move these to client-server but might front-end.

Local PCs & LAN's without mainframe connectivity Very few in retail brokerage.

5. Corporate PC Policy

Who approves PC Purchases Retail Group Management

Who approves PC suppliers Systems Group

Approved PC manufacturer list Unisys/Convergent Technology. Some IBM PCs dial-up.

Who sells and installs the PC's Supplier: ADP

6. Apple Computer Products Installed

Number of Macintoshes installed Almost none...not an approved platform.

Macintosh percentage of total PCs

What departments/applications use Macintoshes A handful
of brokers use Macs at home.

1991 buying plans None

- units/models

- change in Macintosh/PC mix percentageMac down.

- current attitude towards Apple

Not a viable candidate for their next-generation desktop supplier,
mainly due to lack of applications.

7. Using Personal Computers in production applicationsApplications which today use the client/server model

- application description

Broker client data system for 8,000 retail registered reps.

- client description

Unisys/Convergent Technologies Megaframe CTOS server at
each retail office supporting brokers and 3270 data stream to IBM mainframe hosts.

Plans for new applications to use the client/server model

Retail broker order processing, broker prospecting, e-mail, cashier,
interactive new account form.

What client platforms are you considering

Planning to replace 8,000 broker workstations. Can't disclose the three
recommended platforms now under budgetary consideration. Management will choose the entire client-
server technology package as one decision , now pending.

Evaluation of client software environments (subjective comments)

Unable to divulge evaluations

Microsoft Windows 3.0

Hewlett-Packard NewWave

IBM OS/2 Presentation Manager

SCO Open Desktop

Sun Open Look

Apple Computer Macintosh

Other

How will the MIS organization develop expertise in client/server systems development
Develop in-house

Enterprise commitment to IBM's SAA Nothing to
commit to, it seems.

8. Budgets and Sizing Estimates

1990 Systems and Programming Budget \$ (excludes ops) Would not divulge

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions?

Technically far superior to IBM PC, but there are no applications for
brokers running on Macs. Spreadsheets and WP packages are not enough.

Were you aware Apple had such capabilities? No, but it's
the right direction.

Have you considered a front end to existing applications such as the MitemView product I
described?

Yes, we do it.

Would a front end solve a real problem for you? (what, why) Allows multiple
sessions from a single PC.

Which are you likely to invest in

server. - leave existing applications alone They are already client-

Some - front end to existing application such as MitemView

Yes - develop client server model cooperative processing

How much over what time frame
1993

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh?

Yes. However, the GUI advantage Apple has is overblown. A mouse is not a useful tool for a broker.

What do you think of Apple's recent changes in product price/performance
Aware of price cuts. Not applicable to her since Apple is not considered a viable player by her.

Are dealers able to meet your PC/LAN support requirements
ADP devices are serviced by IBM. Bache buys direct from manufacturer and expects nation-wide, uniform service.

What should Apple do to improve its sales to enterprises like yours
1.) Bundle vertical market applications with a pre-configured hardware package.
2.) Work with stock quote vendors: ADP, Reuters, Dow Jones
3.) Would deal direct with Apple and support organization

12. Other notes

A.) Everyone in the brokerage industry has gone with IBM PCs, including Dean Witter, Shearson & Merrill Lynch.

B.) Impression: CTOS incumbent Unisys will not get the workstation replacement contract.

Name/Division Sabre Group, a division of Federated Department Stores

Location 6801 Governor's Island Parkway; Suite 500
Norcross, GA 30071

Type of business All DP for Federated department store chain.
Mission: Applications must be responsive & economical. Sabre is a spin-off of Singer-Harris.

(Telephone) (404) 246-5000

(Size of company) Subsidiary of the Campeau Group

(Major competitors)

2. Respondent Profile

Name Jim Benchler General Systems Manager

Address

Telephone

Responsibilities Applications programming technology, training, and architecture. Evaluating CASE and workstations.

Background and experiences:
21 years in the business. 20 years with Sabre.

Experience with IBM and Apple PCs:
Has some IBM PC experience. Mostly 3270 emulation. PC in his office.

3. Computing Environment

Mainframe Configuration
4 x 3090-600S running ESA. Datacomm DB from Computer Associates (CA). Some DB2. Six versions of IBM DOS/VS. They run 100 CICS regions.

Midrange & Distributed Systems Direction is all mainframe.

Key Applications
1.Credit
2.PO Processing, A/P, Receiving
3.Sales audit and accounting consolidation

4. Terminal Environment

Mainframe
 3270 terminals > 20,000 at the 11 divisions &
 Headquarters.
 3270 active users (local & remote) > 10,000 users + 2,000 devices
 PC LAN users Very little
 Which key applications use 3270's
 All POS systems connect to PS/2 controllers in stores

Local PCs & LAN's without mainframe connectivity
 Many but mostly WP & spreadsheet applications only. No MIS
 coordination.

5. Corporate PC Policy

Who approves PC Purchases CEO

Who approves PC suppliers Benchler & CEO

Approved PC manufacturer list IBM

Who sells and installs the PC's Businessland

6. Apple Computer Products Installed

Number of Macintoshes installed 2 the in building

Macintosh percentage of total PCs 0.1%

What departments/applications use Macintoshes Doesn't know

1991 buying plans

- units/models None
- change in Macintosh/PC mix percentage
- current attitude towards Apple Not visible

7. Using Personal
 Computers in production applications

Applications which today use the client/server model
 None

Plans for new applications to use the client/server model
 No specific plans

What client platforms are you considering
 So much DB checking is required that they see no immediate

applications which would be able to be split into client/server. Advantages of client/ server model generally unclear. High cost to replace 20,000 3270's.

Evaluation of client software environments (subjective comments)

Microsoft Windows 3.0

Hewlett-Packard NewWave

IBM OS/2 Presentation Manager Probable choice. Definitely will use OS/2 for CASE.

SCO Open Desktop

Sun Open Look

Apple Computer Macintosh

Other

How will the MIS organization develop expertise in client/server systems development
In-house over several years.

Enterprise commitment to IBM's SAA None.

8. Budgets and Sizing Estimates

1990 Systems and Programming Budget \$ (excludes ops)
\$25M

1991 S&P budget growth \$ or % up by rate of inflation

1990 Head Count

- Analysts 150
- Applications programmers 224
- Systems programmers, DBAs, network mgr
- PC infocenter
- User training and education

Applications Budget Breakdown (total 100% -- indicate people or \$)

- Maintaining/enhancing existing applications
80%

- Designing/programming new applications 10%

- User training and education 6%

- Systems management and Other 4%

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions?

Would take a lot to get us to look at Apple. Would not consider a pilot project from Apple.

Were you aware Apple had such capabilities?

No.

Have you considered a front end to existing applications such as the MitemView product I described?

Bought a copy of Easel. We are looking at it.

Would a front end solve a real problem for you? (what, why)

Our database checking makes frontware inappropriate.

Which are you likely to invest in

- leave existing applications alone

Probably

- front end to existing application such as MitemViewYes -- Easel

- develop client server model cooperative processing

Maybe

How much over what time frame

Five years.

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh?

Yes

What do you think of Apple's recent changes in product price/performance Doesn't know much about it.

Are dealers able to meet your PC/LAN support requirements Yes.
Businessland support is fine.

What should Apple do to improve its sales to enterprises like yours
It would be very hard to change the long-term relationship we have with IBM. Nothing's going to happen unless Apple convinces top management that they have a better mouse-trap as well as a good reason to do business.

12. Other notes

- A.) MIPS per division are growing as Sabre applications are brought up.
- B.) Looking at CASE but not in place. COBOL/CICS shop
- C.) 15 resident IBM people in S&P alone. Close working relationship.
- D.) IBM PC mandatory for AD/Cycle. Can Apple run compatible CASE tools?
- E.) IBM keeps pushing DB2 but Sabre believes performance is inadequate. IBM is trying to put a square peg in a round hole.
- F.) Computer Associates is trying a service bureau license squeeze-play: pay for CPU license plus a charge for each active user. Sabre has thrown out some CA products and will throw them all out if they have to.

Aberdeen Group and Apple Computer Confidential Page
Aberdeen Front End Survey -- United States Automobile Association Interviewer Peter
Kastner

1. Enterprise Profile

Name/Division	United Service Auto Association
Location	Fredericksburg Rd, San Antonio, TX
Type of business	#6 Insurance company; financial services including 2 banks, mutual funds,
international operations	
(Telephone)	(512) 498-2211
(Size of company)	10,000 users here - almost all are online
(Major competitors)	Allstate, Firemen's Fund

2. Respondent Profile

Name James Dieke Vice President Systems Management

Address

Telephone

Responsibilities

Systems management function including hardware planning, performance and tuning, systems management, executive information. Just transferred to manage all corporate voice and data.

Background and experiences:

4 years here as VP.

Prior: 19 total years at USAA in data processing including systems;
3 years at IBM (unspecified duties)

Experience with IBM and Apple PCs:

Only user in department with Microsoft Windows. Has PS/2 for management. No Apple experience.

3. Computing Environment

Mainframe processor(s) and operating systems (vendor/model/operating system/quantity)

(2) 3090-600S; 3090-400S; 3090-280J; 3090-600J; 3090-380J

Note: 6-30% cycle growth even though business is flat

Midrange & Distributed Systems

4XVAX 8800 cluster for OA E-Mail. Prime, HP, RS/6000 CAD

System/88; (4) Prime; HP 3000; (12) System/36; AS/400

Key Applications

1. Automobile insurance
2. Property & casualty insurance
3. Mutual funds

4. Terminal Environment

Mainframe

3270 terminals	8,100
3270 active users (local & remote)	10,000
PC LAN users	700

Which key applications use 3270's

All customer-related applications are on 3270's. PCs used for standalone decision support and departmental computing.

Local PCs & LAN's without mainframe connectivity
1,100

5. Corporate PC Policy

Who approves PC Purchases Department

Who approves PC suppliers Executive committee

Approved PC manufacturer list IBM. No PC clones. Apple OK.

Who sells and installs the PC's Nynex Business Systems

6. Apple Computer Products Installed

Number of Macintoshes installed Publishing Dept. 20 + scattered individual Macs

Macintosh percentage of total PCs 1%

What departments/applications use Macintoshes Publishing

1991 buying plans

- units/models Doesn't know

- change in Macintosh/PC mix percentage None planned

- current attitude towards Apple Neutral

7. Using Personal Computers in production applications

Applications which today use the client/server model
None

Plans for new applications to use the client/server model

By 1994, USAA plans to rewrite the entire customer service application which is the heart of the business. Cost will exceed \$5M in programming alone -- involves 5,000 client workstations.

What client platforms are you considering

- names PS/2 Model 55SX or up

- reason Runs OS/2 and will run 32-bit OS/2. Corporate dictate to buy IBM PC if PC is required. No clones allowed.

Evaluation of client software environments (subjective comments)

Microsoft Windows 3.0

Hewlett-Packard NewWave

IBM OS/2 Presentation Manager -- USAA-chosen client is PS/2 with OS/2

SCO Open Desktop

Sun Open Look

Apple Computer Macintosh

Other

How will the MIS organization develop expertise in client/server systems development
Boot-strap in-house personnel over time as capability is critical to long-term USAA business. May outsource small pieces.

Enterprise commitment to IBM's SAA
Generally follows, but SAA still not clear to them.

8. Budgets and Sizing Estimates

1990 Systems and Programming Budget \$ (excludes ops) \$20M

1991 S&P budget growth \$ or % Rate of inflation

Applications Budget Breakdown (total 100% -- indicate people or \$)

60%	- Maintaining/enhancing existing applications	
	- Designing/programming new applications	18%
	- User training and education	5%
	- Systems management and Other	17%

Note: The heavy new application budget is caused by extensive work on an imaging system now being deployed, and on the first rewrite of the claims system in 25 years. Client-server not yet a major budget item.

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions?

Apple is a good company, but we usually look to IBM first for these capabilities.

Were you aware Apple had such capabilities?

No -- have never thought of Apple for such a role.

Have you considered a front end to existing applications such as the MitemView product I described?

Still considering client-server software options

Would a front end solve a real problem for you? (what, why)

Idea is clearly intriguing and an eye-opener. Had not considered interim step to client-server. Apple might be able to eliminate 2 desktop terminals at thousands of customer service desks by merging functions now handled separately by an imaging PC and a 3270.

Which are you likely to invest in

- leave existing applications alone Won't do that.

- front end to existing application such as MitemView

Would consider. Particularly liked idea of \$75k pilot using a production application since demos alone can't show the limitations of a technology.

- develop client server model cooperative processing Will do this.

How much over what time frame

4 years for one huge application. No immediate plans for the other numerous applications at USAA.

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh?

Doesn't know.

What do you think of Apple's recent changes in product price/performance

Good.

Are dealers able to meet your PC/LAN support requirements

IBM directed us to buy from Nynex. We have a good relationship with Nynex. We like relationship buying.

What should Apple do to improve its sales to enterprises like yours
The IBM account team is all over USAA. I see occasional Macs on desktops but Apple has no direct marketing presence at USAA that I've seen.

12. Other notes

A.) USAA is the largest employer in the city. Employee turnover is low, so training/retention benefits of workstation and GUI is perceived as low.

B.) While the decision to re-deploy the core customer service application using the client/server model has been made, the decision was hardly unanimous. One vocal camp questions the premise that client-server benefits are worth the costs, and asks why USAA should jump into client-server with the core business application first in an apparent pioneering move.

C.) Due to job change, future USAA contact is Ken Graham, Executive Director of Systems Architecture.

13. Interviewer's Comments

Apple should definitely be calling on USAA in strength.

Aberdeen Group and Apple Computer Confidential
Aberdeen Front End Survey -- University of Texas
Interviewer: Kastner

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1. Enterprise Profile

Name/Division	University of Texas
Location	Austin, TX 78713
Type of business	State University
(Telephone)	(512) 471-7128
(Size of company)	
(Major competitors)	

2. Respondent Profile

Name	Robby (Robert E.) Simpson	Director, DP Division
Randy Ebeling	Asst Director, OS	
Address	Tower Hall	

Telephone(512) 471-0012 (Randy)

Responsibilities

Academic and administrative computing for the university.

Background and experiences:

Experience with PCs

Both are users.

3. Computing Environment

Mainframe processor(s) and operating systems (vendor/model/operating system/quantity)

Grew from an IBM 3090 model 200 to 300E to 400S to 720 in less than 5 years. Runs MVS...expects no problems in going to ESA. Gets 50% discount from IBM on hardware purchases. MIPS consumption growing at 50% per year -- MIPS are replacing clerical & accounting personnel. Amdahl has all other Texas state business.

Midrange & Distributed Systems

None

Key Applications

1. All University administrative applications
2. All University academic applications
3. In 1991, will eliminate paper from most business procedures.

4. Terminal Environment

Mainframe	4,500 total
3270 terminals	<60%
3270 active users (local & remote)	
PC LAN users	with 3270 emulation >40%
Which key applications use 3270's	All applications are host-based

Local PCs & LAN's without mainframe connectivity

5. Corporate PC Policy

Who approves PC Purchases No controls

Who approves PC suppliers MIS supports PCs & Macs. Sun & DEC not supported.

Approved PC manufacturer list PC/Mac suggested over 3270. IBM is offering distributor status as a way to break Apple's hold.

Who sells and installs the PC's MIS installs admin & academic PCs. MIS runs Infocenter for student PCs.

6. Apple Computer Products Installed

18,000 sold to date; includes students who have graduated.

Number of Macintoshes installed

Largest Mac site in U.S. 2,000 in administration alone.

Macintosh percentage of total PCs

What departments/applications use Macintoshes

Spread everywhere - usually a personal, not department choice.

1991 buying plans

- units/models

- change in Macintosh/PC mix percentage

- current attitude towards Apple

Very High acceptance rate anticipated on Apple program to trade in and trade up to Mac. Will accept PCs as well as older Macs.

7. Using Personal Computers in production applications

Applications which today use the client/server model

None. They do support downloading budget information.

- application description

- client description

Plans for new applications to use the client/server model

What client platforms are you considering

- names

- reason

Evaluation of client software environments (subjective comments) They write applications to 3270's. Up to client to handle front end.

Microsoft Windows 3.0

Hewlett-Packard NewWave

IBM OS/2 Presentation Manager

SCO Open Desktop

Sun Open Look

Apple Computer Macintosh

Other

How will the MIS organization develop expertise in client/server systems development
Works closely with suppliers.

Enterprise commitment to IBM's SAA

8. Budgets and Sizing Estimates

Runs on a P&L basis even though University is non-profit.

1990 Systems and Programming Budget \$ (excludes ops) \$2M

1991 S&P budget growth \$ or % Operating budget flat.
Capital budget up slightly.

1990 Head Count

- Analysts 75 central P/A's + 200 P/A's in user departments + 250 user "programmers"
- Applications programmers
- Systems programmers, DBAs, network mgr
- PC infocenter
- User training and education

Applications Budget Breakdown (total 100% -- indicate people or \$)

- by
 - Maintaining/enhancing existing applications 10-15% - low design
 - Designing/programming new applications 65% - "our real job"
 - User training and education 20% - high by design
 - Systems management and Other %

9. Describe Apple Computer's capabilities for terminal emulation, front end, and cooperative processing. Focus on MitemView and the concept of non-invasive front ends.

10. Impressions of Apple's Front End Capabilities

What are your first impressions?

Strongly believes front-ending is desirable. Actively implements front-ending using Software AG's (SAG) Natural Connection on Macs. Does not believe there is substantial benefit to cooperative processing today.

Were you aware Apple had such capabilities? Yes.

Have you considered a front end to existing applications such as the MitemView product I described?

Yes -- Use SAG. Investigated MitemView but it is too complex. Had some problems with a library system. Contact Dave Koehler, DP Director, of Cornell's Mandarin project. Thinks Mitem was written by techies with little thought about the user interface.

Would a front end solve a real problem for you? (what, why)

Which are you likely to invest in

- leave existing applications alone
No
- front end to existing application such as MitemView Yes, but not MitemView
- develop client server model cooperative processing
Maybe in a couple of years when tools are more mature.

How much over what time frame

The University is using front-ends on Macintosh today.

11. Collateral Issues

Are IBM-compatible PCs catching up to Macintosh

Not here. Users are offered character-made applications on PCs or GUI-applications on Mac.

What do you think of Apple's recent changes in product price/performance

Latest prices, coupled with used equipment buyback, are winners. Current order backlog a problem.

Are dealers able to meet your PC/LAN support requirements

IBM is trying to make the University a "dealer" in order to counter the attractive Apple educational discount.

What should Apple do to improve its sales to enterprises like yours

We enjoy a close working relationship with Apple which has enabled us to achieve notable successes. Without this relationship, we would probably do less with Macintosh.

12. Other notes

A.) Believes cooperative processing can only occur with products like Software AG's Natural Connection on PC or Mac. Without tools like Natural, coding is too complex.

B.) Believes a Mac server such as Avatar to the IBM host is easier and less costly than multiple 3270 links to the host via PCs.

Slide 1 - Introduction

Aberdeen research shows considerable interest in cooperative processing

Cooperative processing means rewriting applications

Question is "How will we get there?"

Slide 2 - Existing Systems

Massive installed base...not easily changed...will take time

Data hard to get at for decision support

3270 or equivalent requires extensive operator training

Getting data to desktop is fairly involved

Slide 3 - The Challenge

Is there an interim step between old terminals/emulation and cooperative processing in mid-1990s applications?

Yes there is.

We call this interim step front-ending

Slide 4 - The Vision

Gain the benefits of modern desktop workstation tools which are graphically-oriented, easy to use and learn, and tied to personal applications

Keep the existing host applications intact and unchanged...
the cost of changing them is too high

Slide 5 - Extending Host Systems

Front ends simplify access to the host and control host access

Intelligent workstations will be the clients of the future, so the hardware investment is protected during the transition

Front-ends can improve data use by making it readily available to workstation applications like spreadsheets

Slide 6 - Non-Invasive Front-Ending

Non-invasive means the host never knows about the front-end client...the host thinks it has an expert user

Zero additional cost to enhance/upgrade/maintain host side of front-end

Workstation to host connections are over existing
3270 or async connections...including remote lines

Slide 7 - Client & Product

Our client is Apple Computer.

Apple has asked Aberdeen to survey users like you to determine your thoughts about front-ending ... and Apple's role in the enterprise systems market

Mitem Corp's MitemView is the Apple co-marketed product best suited to robust front-ending on Macintosh

Apple also has a program with a major systems integrator to assist enterprises in piloting front-ends with a technology transfer and on-site development of a prototype application

Slide 8 - Apple's Host Connectivity Products

You may be unaware of the extent of Apple's connectivity to hosts:

- Terminal emulation includes IBM, DEC and HP
- Front-ends include MitemView and MacProfs
- Cooperative processing tools include:
- Data Access Language interface to RDBMSs
- MacWorkstation - a Macintosh client
- MacAPPC software for LU 6.2

Metadata 2024

1. Concise Summary of the Findings in the Document

The document is a comprehensive report by Aberdeen Group for Apple Computer USA, focusing on the market for frontware and client-server computing in the early 1990s. It includes detailed findings from interviews with major enterprises across various industries, highlighting their computing environments, corporate PC policies, and attitudes towards Apple Computer's products. The report identifies a significant market resistance to Apple products in favor of IBM PCs, driven by entrenched corporate policies and misconceptions about Apple's capabilities. It also discusses the potential for Apple's frontware solutions, such as MitemView, to serve as non-invasive front-ends to mainframe applications, and the broader implications for Apple's market strategy.

2. Summary of How Computer Technology Buyers Should React to the Document

Computer technology buyers should recognize the potential benefits of Apple's frontware solutions for enhancing productivity and reducing mainframe load through non-invasive front-ends. The document argues that buyers should be open to evaluating Apple's capabilities beyond the entrenched IBM PC architecture, particularly in terms of graphical user interfaces and connectivity to mainframes and LANs. Buyers are encouraged to consider pilot programs to test Apple's solutions in real-world applications, which could lead to significant long-term benefits and cost savings.

3. Analysis on the Technology Discussed and Its Relation to Mainstream Computing in 2024

The technology discussed in the document, particularly frontware and client-server computing, laid the groundwork for modern distributed computing environments. In 2024, the principles of non-invasive front-ends and client-server models have evolved into cloud computing and microservices architectures, which dominate the enterprise IT landscape. The emphasis on graphical user interfaces and seamless connectivity has become standard, with modern applications leveraging web-based interfaces and APIs to integrate disparate systems. The document's insights into the early adoption challenges and market resistance provide valuable historical context for understanding the evolution of enterprise computing.

4. Analysis on How the Contents of the Document Played a Role in Impacting the History of the Companies Mentioned

The document highlights the significant market challenges Apple faced in penetrating the enterprise market dominated by IBM. Despite these challenges, Apple's persistence in promoting its graphical user interfaces and connectivity solutions eventually paid off, contributing to the company's resurgence in the late 1990s and early 2000s. The insights from the report likely influenced Apple's strategic decisions to focus on consumer markets initially, before making a successful return to the enterprise space with products like the iPhone and iPad, which have become integral to modern business environments. The document also underscores the importance of targeted marketing and direct engagement with potential buyers, lessons that have shaped Apple's approach to enterprise sales.

5. Market Acceleration Catalysts

- Increased awareness and education about Apple's enterprise capabilities.
- Successful pilot programs demonstrating the benefits of Apple's frontware solutions.
- Strategic partnerships with systems integrators and software vendors.
- Competitive pricing and improved support services for enterprise customers.
- Enhanced marketing efforts targeting key decision-makers in large enterprises.

6. Keywords

frontware, client-server computing, Apple Computer, IBM PC, enterprise IT, graphical user interface, mainframe connectivity, non-invasive front-ends, market resistance, pilot programs

Citations:

[1] <https://ppl-ai-file-upload.s3.amazonaws.com/web/direct-files/18598228/6a9627be-c84d-4514-bb97-83b60f55a395/APPLERPT.pdf>

[2]
https://archive.org/stream/susquehannaalumn5560susq/susquehannaalumn5560susq_djvu.txt

[3] https://en.wikipedia.org/wiki/Peter_Kastner

[4]
https://archive.org/stream/computerworld3247unse/computerworld3247unse_djvu.txt

[5] <https://archive.org/details/appleconfidentia00linz>

[6] <https://www.ciotalknetwork.com/contributor/peter-kastner/>

[7] <https://www.cnn.com/TECH/computing/9903/29/apple.confid.idg/index.html>

[8] <https://www.linkedin.com/in/peter-miller-06400039>

[9] https://books.google.com/books/about/Apple_Confidential.html?id=3KG1AAAAIAAJ

[10] <https://dl.acm.org/doi/10.5555/520621>

[11] https://www.goodreads.com/book/show/1908485.Apple_Confidential